CAPSTONE PROJECT

NUTRIMIND: AN LLM NUTRITION ASSISTANT PROTOTYPE

Presented By:-

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OUTLINE

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PROBLEM STATEMENT

- **The Challenge: Scaling Expert-Level Personalized Nutrition**

In the rapidly growing digital health market, personalized nutrition is a core consumer demand. However, most digital tools offer only generic, one-size-fits-all meal plans. This approach fails to account for an individual's unique health goals, dietary restrictions, and medical conditions, leading to low user engagement and poor health outcomes. This creates a significant industry challenge: providing scalable, expert-level nutrition counseling that is both affordable and deeply personal. The crucial part is the development of an intelligent AI agent that can reason over large-scale dietary databases to generate dynamic, personalized meal plans and adaptive advice, effectively acting as a virtual nutrition expert for every user.

PROPOSED SOLUTION

The proposed system, "NutriMind," aims to address the challenge of providing scalable, expert-level personalized nutrition guidance. This involves using an agentic AI architecture on the IBM watsonx platform to generate personalized meal plans. The solution consists of the following components:

- Data Collection:
  - Sourced the "Daily Food and Nutrition Dataset" from Kaggle to serve as the knowledge base (later found to be synthetic; see Limitations).
- Data Preprocessing:
  - Performed manual data cleanup using Microsoft Excel to remove duplicates (this step later turned out to misalign rows; see Limitations).
- Retrieval-Augmented Generation:
  - Implemented a RAG pipeline to ground the AI's answers in retrieved dataset records.
  - Used the granite-embedding-278m-multilingual model to convert the cleaned dataset into a vector index for retrieval.
- Prompt Engineering & Prototyping:
  - Leveraged the IBM watsonx Prompt Lab for rapid prototyping of the agent's core skills.
  - Engineered and tested prompts for meal plan generation and food swap recommendations against the granite-3-8b-instruct model and iterated on their outputs.

PROPOSED SOLUTION

- **Agentic AI Architecture:**
  - **Developed the core "NutriMind" agent using the IBM watsonx Agent Lab, leveraging the LangGraph framework to orchestrate the tool-calling loop.**
  - **Implemented a ReAct (Reason and Act) architecture so the model decides when to call which tool.**
  - **Integrated the llama-3-3-70b-instruct model as the primary reasoning engine for understanding complex user queries.**
  - **Equipped the agent with a multi-tool system, including the custom nutritional RAG database, Google Search, Wikipedia Search, and a WebCrawler for questions outside the dataset**
- **Deployment:**
  - **Tested the agent through the watsonx platform's built-in preview chat.**
  - **Deployed the final solution to a watsonx deployment space on the IBM Cloud platform (Lite tier), which exposes an API endpoint; the deployment is no longer live.**
- **Evaluation:**
  - **Assessed the agent's performance qualitatively by manually testing a set of nutrition queries; no quantitative metrics were collected.**
- **Result**
  - **The agent generated custom meal plans for user-specific requests and provided context-aware suggestions on how to replace certain food items with healthier alternatives. Nutritional figures were not reliably accurate (see Limitations).**

SYSTEM APPROACH

The "System Approach" section outlines the overall strategy and methodology for developing and implementing the NutriMind AI system.

- System Architecture:
  - **Cloud Platform:** IBM Cloud (Lite Tier)
  - **AI Environment:** IBM watsonx.ai Studio
  - **Core AI Components:**
    - **IBM watsonx Agent Lab:** For final agent assembly, tool integration, and deployment.
    - **IBM watsonx Prompt Lab:** For rapid prototyping, prompt engineering, and RAG pipeline development.
- Technology & Models Used:
  - **Reasoning Engine (LLM):** llama-3-3-70b-instruct
  - **Prototyping Model (LLM):** granite-3-8b-instruct
  - **Embedding Model (for RAG):** granite-embedding-278m-multilingual
  - **Agent Framework:** LangGraph
  - **Agent Architecture:** ReAct (Reason and Act)
  - **Data Pre-processing Tool**: Microsoft Excel

ALGORITHM & DEPLOYMENT

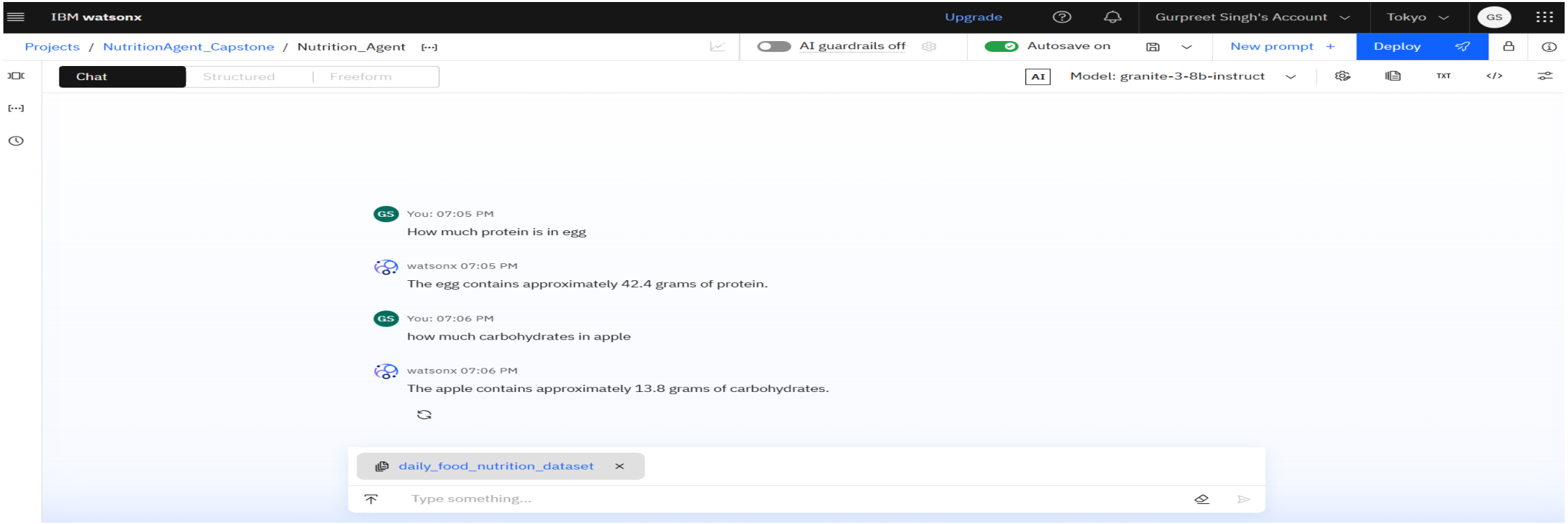
- In the Algorithm section, we describe the AI architecture chosen for generating personalized nutritional guidance. Here's the structure for this section:
- **Algorithm Selection:**
  - **A ReAct (Reason and Act)** agentic architecture was implemented, powered by **the llama-3-3-70b-instruct model**. This approach was justified by the need to handle complex, multi-step user queries that require both internal knowledge retrieval and external tool use, which a single-turn chatbot without tools cannot handle.
- **Data Input:**
  - The primary input is the **user's natural language query** (e.g., "create a meal plan for me").
  - The secondary data source is a vectorized nutritional database, created from a pre-processed Kaggle dataset using the **granite-embedding-278m-multilingual model**. This serves as the agent's internal knowledge base.
- **Prompt Development (no model training):**
  - The agent's core "skills" were developed through a process of prompt engineering in the **IBM watsonx Prompt Lab**. Specific, detailed prompts for meal plan generation and food swap recommendations were prototyped and tested against the **granite-3-8b-instruct model.**
- **Prediction Process:**
  - The agent makes "predictions" (generates responses) by first using the **ReAct framework** to reason about the user's query. It then autonomously selects the appropriate tool—either querying the **internal RAG database** for nutritional facts or using external tools like Google Search for current information—before synthesizing the retrieved data into a final answer.

RESULT

The results of the "NutriMind" AI agent in generating personalized nutritional guidance. Included are visualizations from the capstone demonstration and comparisons between a user's complex query and the agent's tailored output to illustrate the model's behaviour, including its limitations.

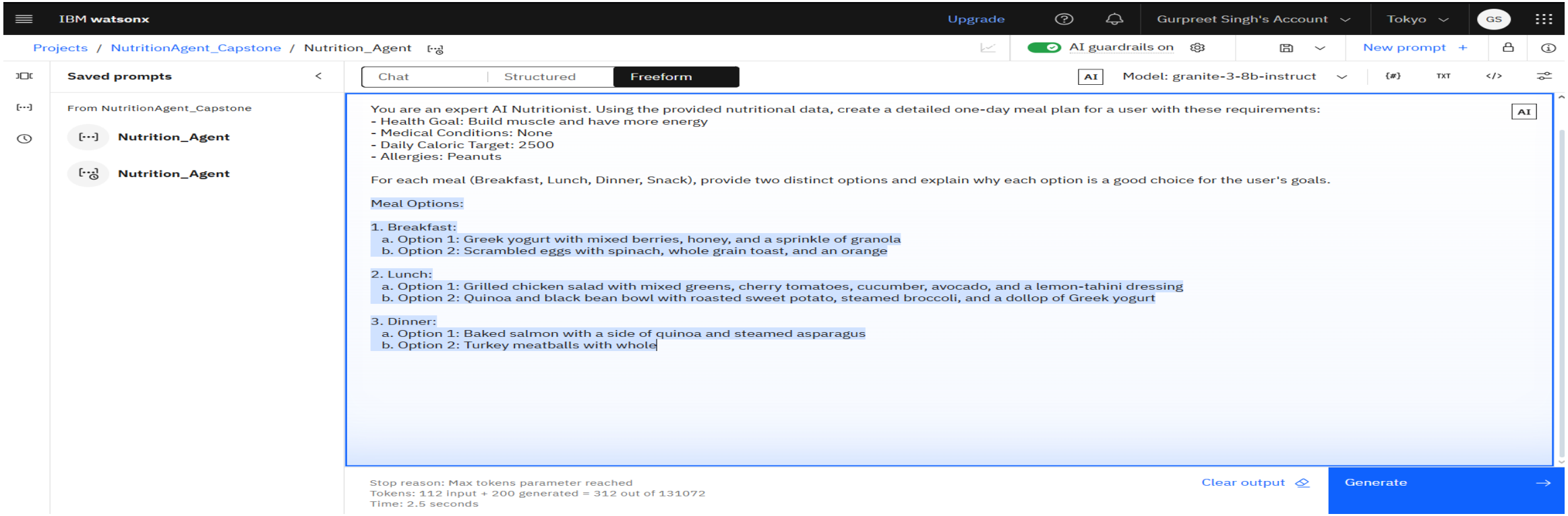
RESULT

- This screenshot shows a test of the Retrieval-Augmented Generation (RAG) pipeline within the **IBM watsonx Prompt Lab**. The system, grounded with the daily_food_nutrition_dataset, is being queried for specific nutritional facts. Using the granite-3-8b-instruct model, the model returns values taken directly from the dataset, which confirms that retrieval works. Note: the dataset value for eggs (42.4 g protein) is not realistic (a real egg has about 6 g), so grounding only helps when the indexed data is correct.



RESULT

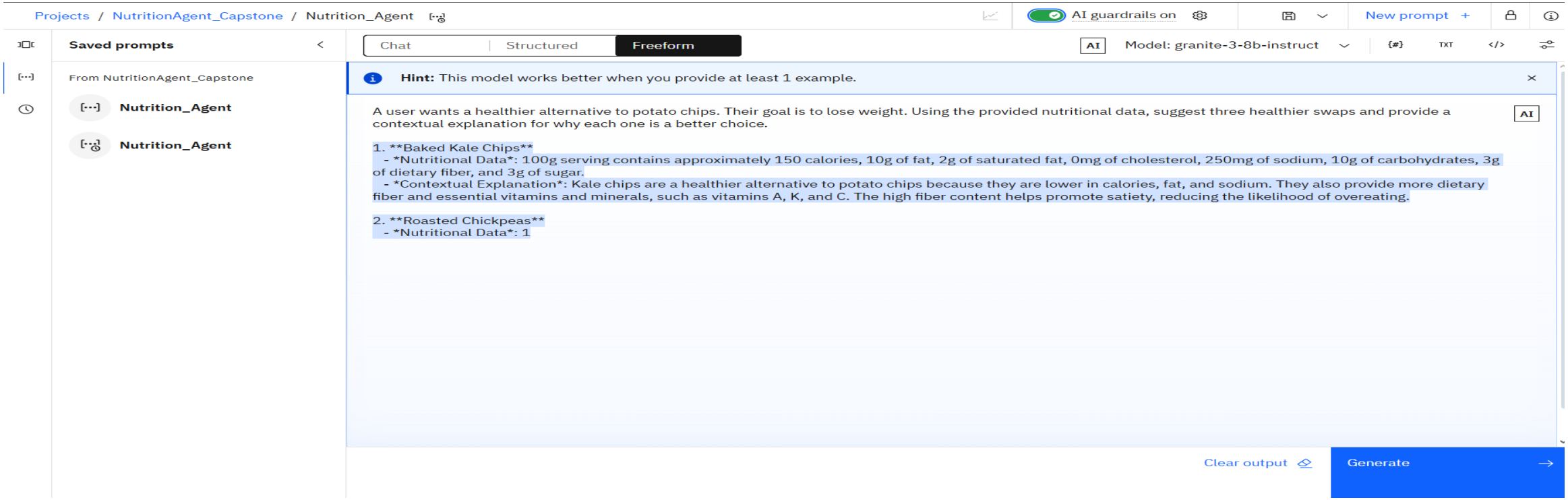
This screenshot captures the generation of a personalized meal plan in the **Prompt Lab**. The granite-3-8b-instruct model was given a persona and a user request with multiple constraints (Health Goal: Build muscle, 2500 Calorie Target, Peanut Allergy). The model generated a structured, multi-option meal plan for Breakfast, Lunch, and Dinner; the output stopped at the 200-token limit before finishing.



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RESULT

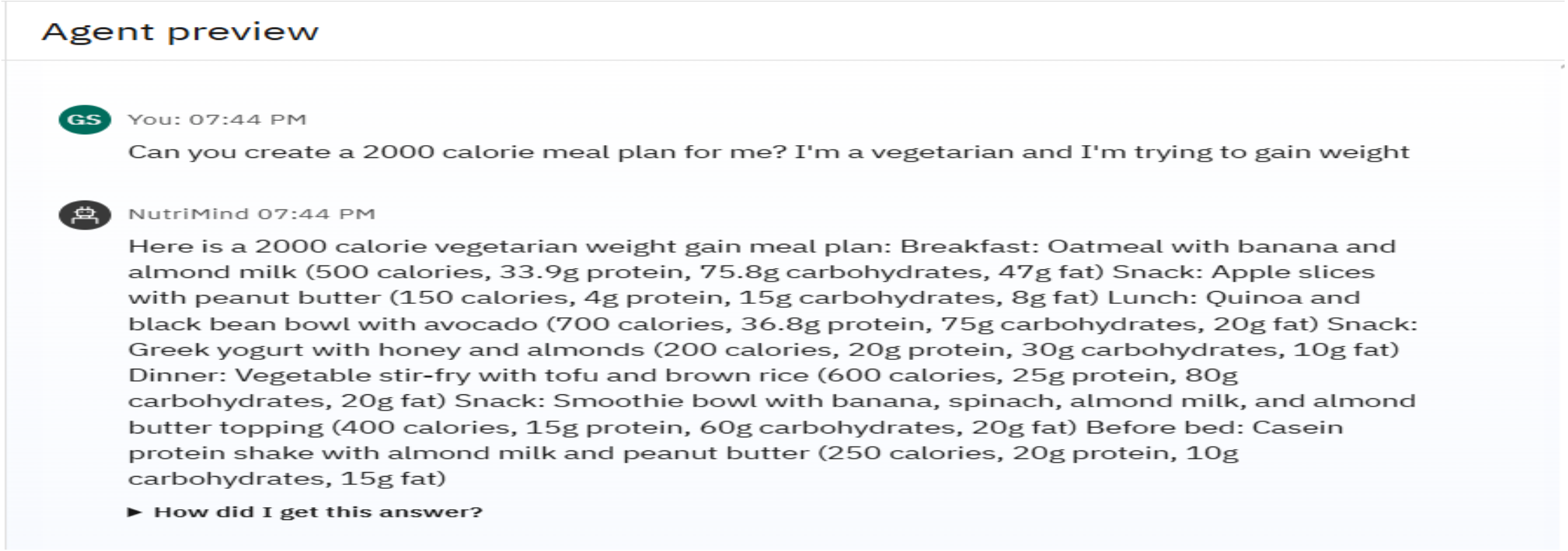
This image showcases the **IBM watsonx Prompt Lab** in "Freeform" mode, where a core agent skill was prototyped. A prompt was provided to the **granite-3-8b-instruct** model, tasking it with suggesting and explaining healthier food swaps based on a user's goal. The model generated alternatives like "Baked Kale Chips" with explanations; the output was cut off at the token limit, and its nutrient figures were not checked against a reference source.



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RESULT

This image displays a test of the assembled **"NutriMind"** agent in the **IBM watsonx Agent Lab preview**. The query: a 2000-calorie, vegetarian, weight-gain meal plan. The agent interpreted the user's intent and generated a structured vegetarian plan, but the listed items add up to about 2,800 kcal, not 2,000, so LLM arithmetic needs a deterministic check.



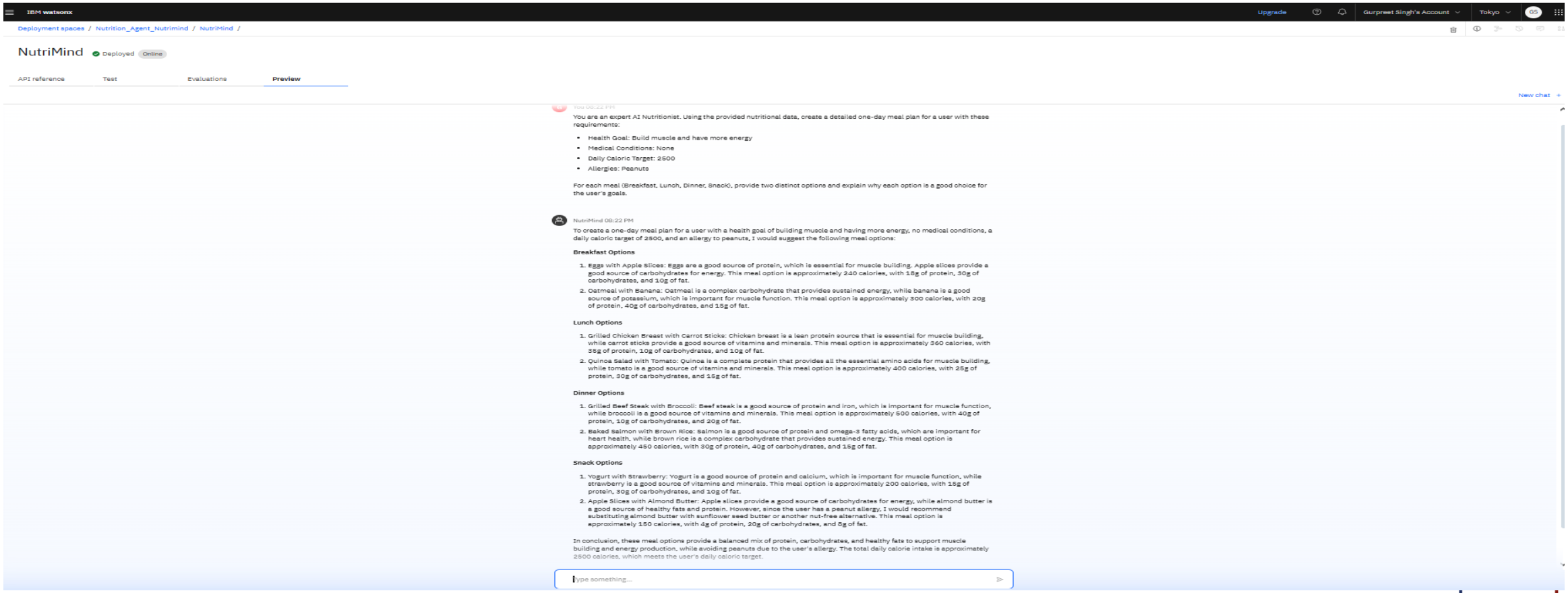
RESULT

This screenshot demonstrates the agent's ability to provide context-aware nutritional advice. When asked for a low-sugar replacement for soda, the "NutriMind" agent suggested alternatives. The response recommends water, unsweetened tea, and seltzer, and adds context by advising against high-sugar fruit juices, which is sensible general advice.



RESULT

This image captures a test of the deployed "NutriMind" AI service. The screenshot shows the deployment's "Preview" interface while the deployment was running on IBM Cloud (Lite tier). The agent processed a natural-language request for a personalized meal plan, and returned a structured, multi-option plan that avoids peanuts. However, the stated daily total (about 2,500 kcal) does not match the listed options, which add up to about 1,200-1,400 kcal per day.



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CONCLUSION

- Developed a proof-of-concept AI Nutrition Assistant, **"NutriMind,"** on the **IBM watsonx platform.**
- Demonstrated a multi-tool, RAG-powered agentic architecture that produces personalized, context-aware nutrition suggestions.
- The project covered the full prototype workflow—from data preparation and prompt engineering to the final assembly and deployment of an agent. The main lesson: answer quality is limited by data quality and by unchecked LLM arithmetic (see Limitations).

LIMITATIONS & LESSONS LEARNED

- Source data: the Kaggle dataset appears synthetic. In about 58% of rows, the stated calories differ by more than 50% from the standard 4/4/9 kcal-per-gram estimate from protein, carbs and fat.
- Data cleanup: de-duplicating the food-name column in Excel shifted food names out of line with their nutrient values in the cleaned file.
- RAG retrieved these values faithfully (e.g. 'egg: 42.4 g protein'; a real egg has about 6 g). Grounding is only as good as the indexed data.
- LLM arithmetic: listed meal totals did not match the calorie targets (about 2,800 kcal for a 2,000 kcal request).
- Evaluation was qualitative only; there were no automated tests or metrics.
- Lessons: validate data before indexing, use deterministic tools for calculations, and evaluate against a fixed set of test queries.

FUTURE SCOPE

- **Multimodal Input Integration:**
  - **Image Recognition:** Integrate **IBM Watson Visual Recognition** to allow users to upload photos of food items or grocery labels. The service would identify the food, and its name would be passed as text to the NutriMind agent for analysis and recommendations.
  - **Voice Commands:** Incorporate **IBM Watson Speech-to-Text** to enable a hands-free, conversational experience, allowing users to ask questions and receive nutritional advice via voice.
- **Dynamic Feedback & Personalization Loop:**
  - Implement a user profile database to store individual preferences, feedback, and dietary history.
  - The agent would query this database before generating recommendations to ensure it learns and adapts over time (e.g., "remembering" that a user dislikes a certain food).
- **Advanced Health Data Integration:**
  - Connect the agent to health and fitness APIs (like from smartwatches or fitness trackers) to automatically pull in data on activity levels and caloric expenditure, leading to even more precise and dynamic meal planning.

REFERENCES

- **Dataset:**
  - Shamim, A. (2023). *Daily Food and Nutrition Dataset*. Kaggle. Retrieved August 4, 2025, from https://www.kaggle.com/datasets/adilshamim8/daily-food-and-nutrition-dataset
- **Core AI Methodologies:**
  - Lewis, P., et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. This foundational paper introduced the RAG technique used to ground the agent in retrieved dataset records. Link: https://arxiv.org/abs/2005.11401
  - Yao, S., et al. (2023). *ReAct: Synergizing Reasoning and Acting in Language Models*. This paper details the ReAct architecture, which was implemented to enable the agent to reason about a user's query and intelligently select the correct tool.Link: https://arxiv.org/abs/2210.03629
- **Platform & Technology Documentation:**
  - IBM. (2025). *IBM watsonx.ai Documentation*. This was the primary technical resource for developing and deploying the agent on the IBM Cloud platform. Retrieved August 4, 2025, from https://cloud.ibm.com/docs/watsonx-ai.

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